

# Human-Robot Scaffolding: An Architecture to Foster Problem-solving Skills

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In order to give assertive support, robots need to understand the cognitive and emotional characteristics of learners while in the learning process. Also, if the task involves handling capacity, robots must have similar skills. Three concepts were explored to control the cognitive and emotional robot's behavior: the psychological flow theory, the scaffolding pedagogic strategy, and the multiagents-software paradigm. Based on these concepts, the Human-Robot Scaffolding architecture was designed. It is divided into five blocks. First, the sensory block recognizes body gestures, speech, and task state. Second, the beliefs block estimates the skills and emotional state of learners. Third, the desires block validates the goals the robot can reach; the goals are grouped in skills development, emotional control, cognitive control, challenge control, life signals, and immediate support. Fourth, the intentions block, based on the goals competition strategy, selects the goal that the robot will perform. Finally, the action-planner block regulates the robot's movements according to the robot's emotions. The validation procedure was done with 53 learners ranging between 10 and 13 years old who study in public and private schools. Based on the research achievements, the robot fosters learning the Mean-Ends Analysis strategy and the solution of a problem. A video fragment that summarizes the research process is available in <https://youtu.be/qbohCjBlwYc>.

CCS Concepts: • **Applied computing** → **Interactive learning environments**; • **Computer systems organization** → **Robotic autonomy**;

Additional Key Words and Phrases: Anthropomorphic robots, human-robot scaffolding, multi-agent systems, robots for education

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## 1 INTRODUCTION

The assertive intervention of robots during the learning process is one of the challenges of HRI. Assertive intervention means taking actions according to the cognitive and emotional needs of the subjects [57]. The actions done by robots to attend cognitive needs are evidenced in [15, 18]. Additionally, the actions to support the emotional needs are evidenced in [17, 29, 39, 63]. Although there are various studies about the educational actions of robots [15, 33, 48], there are opportunities for studying the conditions to be assertive. For instance, the actions have to merge cognitive,

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emotional, and physical variables while robots grasp objects. In learning environments, robots are used such as tools with basic principles of movement, automatic tools, devices with deliberative behavior, tools with emotional and cognitive characteristics, and tools with social attributes [69, 74]. The advances in social robots with educative purposes are being adjusted to three aspects: cognitive convergence, emotional convergence, and handling capability. Developing cognitive convergence has been a goal in HRI. This goal aims to integrate different information resources to ensure that the robot understands the subject's communicative act. Thus, during the interaction between robots and humans, it is intended to take advantage of explicit communication resources such as speech and implicit ones such as body language. In the evolution of multimodal fusion methods, some works are highlighted. For instance, the development of an architecture based on multiagent systems to organize the information from various sensors during cooperative mechanical assembly [38], a multimodal instruction system that integrates natural language and artificial vision [37], the JAST architecture that integrates different user communication channels to interact with a robot composed for two arms and the robot iCat [25], the second JAST version that recognizes objects; speech; and tracking the face, gaze, and hand movements to interpret the model of the world to interact with the user during the assembly of blocks [61], the *Haptic-Ostensive Reference* architecture to recognize the *deictic* movements during the cooperative process [24], the *multimodal fusion* method that integrates speech, object recognition, and gestures for the design of a symbolic inference system to detect successes and errors, of the verbal and non-verbal behaviors of the user, to propose solution strategies during the mechanical assembly process [26]. Different authors have developed architectures merging psychology, pedagogic and didactic principles which were proposed by Jerome Bruner [77], and Roy D. Pea [56]. For instance, Mary Foster developed a theoretical approach to the scaffolding strategy using robots, [23, 58]. Currently, Brian Scassellati is studying conditions to personalize the robot behavior with educative purposes, [70], and the SIRP research group at the Javeriana University is developing robotic platforms based on the theatrical principles to foster learning in non-technological areas [9].

The use of anthropomorphic robots with educational purposes is recent. Because robots are considered tools to shape thinking [65], robots are used as learning objects and collaborative agents [24, 36, 59]. As collaborative agents, social robots are a novel expression of robots in education [3]. The robot's design has evolved to adapt their cognitive, emotional, and social behavior based on the learner's characteristics [2]. The anthropomorphic, cognitive, and emotional characteristics have generated interest in the educative field. The integration of these characteristics considers the convergence of aspects cognitive [41], emotional [30, 31], and physical [23, 52]. The design of architectures should then harmonize these aspects through the scaffolding strategy through roles as tutor, peer, and learner. As tutors, robots need a balance between functionality and expressivity. Functionality considers the technological characteristics to support the learning task through physical, cognitive, and emotional mediation [28, 42, 46]. Expressivity has phenomenological implications that affect the learner's behavior through confidence and mood, [45]. As peers, robots influence learning through imitative learning and following the problem-solving process by the more skilled peer, [5, 51]. As learners, robots benefit learning through behaviors linked to motivation, creativity, curiosity, and attention, [55, 67]. Generally, educative research uses robots with human, animal, or monster appearance, generally without manipulative capacity, but with emotional transmission capability. With these resources, aspects such as: relationship between physical presence and perception of support have been evaluated [4, 44], emotion transmission mechanisms [40], aspects of the robot's non-verbal language [73], non-verbal behavior of users [35] and verbal interventions of robots to promote thinking actions in users [36]. For instance, in search of emotional communication during the development of the activity, the robots' behavior includes signs of life, gaze tracking, a protocol for starting the task, use of name or learner, and tone of voice [5].

As collaborative agents in learning environments, the HRI phenomenon needs to be studied. The convergence of cognitive, emotional, and physical demands acknowledges the robot's influence to achieve the learning goals. Although social robots are designed to guide the learning process, there are phenomenological issues that influence student performance. Some instruments to evaluate these HRI features are presented below: *Anxiety: Robot Anxiety Scale (RAS)* [49, 50]; *Negative Attitudes towards Robots Scale (NARS)* [72]; *Imagined Contact (IC)* [78]; *Educational Robot Attitude Scale (ERAS)* [66]. *Attitude: Negative Attitudes towards Robots Scale (NARS)* [72]; *The Unified Theory of Acceptance and Use of Technology (UTAUT)* [32]; *The Technology Acceptance Model (TAM)* [21]. *Testing the Technology-to-Performance Chain Model (TPC)* [68]; *Positive and Negative Affect Schedule (PANAS)* [75]. *Cognitive Convergence: Psychological Scale for General Impressions of Humanoids (PSGIH)* [34] *Likeability Scale.* [60]. *The Lombard and Ditton's Engagement Scales* [43]; *SHAPE Automation Trust Index (SATI)* [1]. *Usability: Multi-dimensional Assessment of telepresence RoboT for older Adults (MARTA)* [79], *Questionnaire for user interface satisfaction (QUIS)* [16]. *Sociability: Activity for Daily Living (ADL), Instrumental Activities of Daily Living (IADL)*[10]. *Social Influence: Robotic Social Attributes Scale (RoSAS)* [53]. *Perception of Intelligence: Trust Perception Scale-HRI* [64]. *Empathy: Robot's Perceived Empathy (RoPE)* [14]. *Anthropomorphism: Individual Differences in Anthropomorphism Questionnaire (IDAQ)* [76], *Rasch Model* [6].

This document proposed the design of the **Human-Robot Scaffolding architecture (HRS-EDU)** based on two features to regulate the robot's behavior. First, the flow theory evaluates if a learner is focused, bored or anxious. The focus state rises because of an equivalent relation between the learner's skills and the problem's challenge. The boredom state rises when the learner's skills are more significant than the basic skills to solve the problem. The anxiety state rises when the learner's skills are lower than the skills required to solve the problem [20]. Second, the scaffolding strategy fosters the learning process according to the learner's skills to solve the problem. The scaffolding suggests that the agent with more skills has the teaching responsibility, but it decreases as fast as the learner develops skills. The transition of responsibility takes place through four types of mediation: focus lesson, guided instruction, collaborative work, and independent work, [22]. In conclusion, the contribution of this paper is a model of cognitive architecture that regulates the robot's behavior based on the scaffolding strategy and the practical reasoning theory that explains the agent's behavior through his beliefs, desires, and intentions. The following sections explain the design of the HRS-EDU architecture based on the theory of BDI agents (*Beliefs, Desire, Intentions*).

## 2 DESIGN OF THE ARCHITECTURE HRS-EDU

HRS-EDU supports the learning process while learners solve a well-defined problem. The support considers the scaffolding strategy and the flow theory. The robot behavior is also motivated by the BDI multiagents paradigm grounded on the practical reasoning theory that explains human behavior according to beliefs, desires, and intentions [8]. Thus, the architecture has four modules: *Sensorial, Beliefs, Desires-Intentions, Action*. See Figure 1.

### 2.1 Sensorial Module

As shown in Figure 1, HRS-EDU uses three sensors to recognize the learner's movements, facial expressions, and speech. The architecture uses a microphone  $S_1$  to recognize voices and two depth cameras  $S_2, S_3$  to recognize the emotional gestures, body language, and task state. The array  $S$  shows the data:

$$S = [S_5, S_6, S_7, S_8, S_9], \quad (1)$$

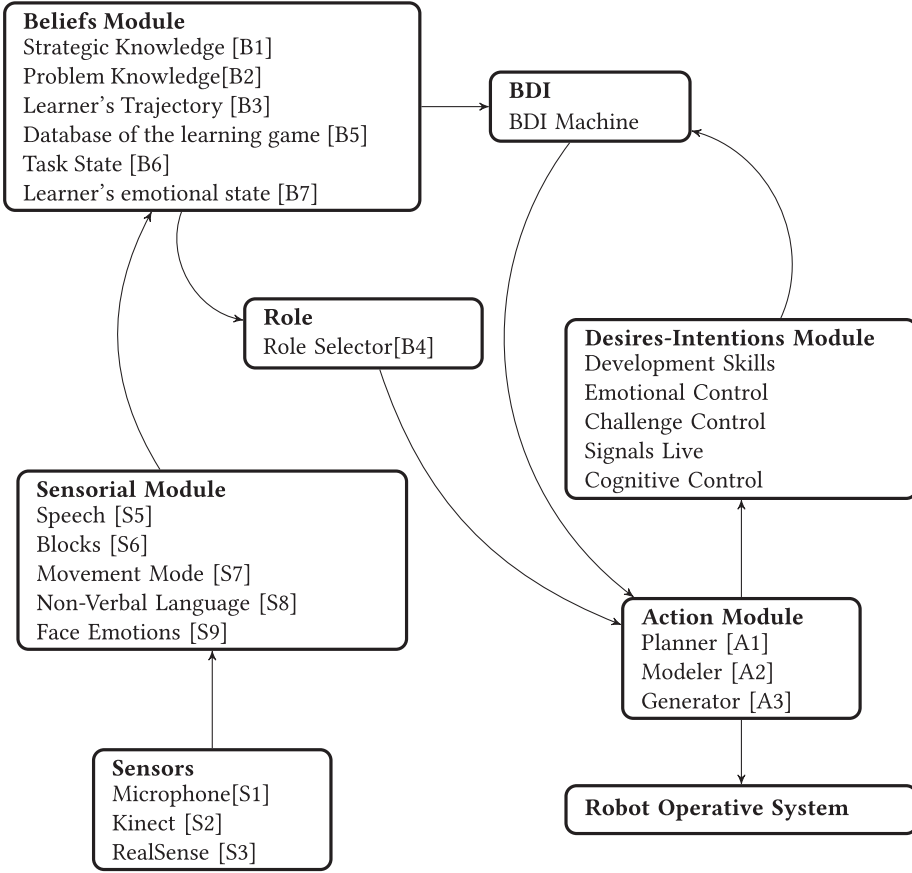


Fig. 1. Human-Robot Scaffolding Architecture *HRS-EDU*. The practical reasoning theory is the first level of *HRS-EDU*. Based on this theory, the believes, desires, and intentions modules were proposed. The beliefs module depends on the cognitive and emotional behavior of learners during the problem-solving process. The desires and intentions modules were designed based on both the pedagogical scaffolding theory and the psychological flow theory. Finally, the actions of the robot were designed according to the capabilities of the Baxter robot.

where:

- $[S_5] \rightarrow$  **Words**. Six words were proposed:  $[S_5] = [Baxter\ Help\ me!,\ Yes,\ I\ get\ it,\ No,\ I\ don't\ understand,\ Aha!]$ . When  $s_{5_i} \rightarrow [True, False]$ .
- $[S_6] \rightarrow$  **Blocks**. With a deep camera, the block's recognition process was done. The blocks were arranged on a bi-dimensional space with  $8 \times 4$  positions. This arrangement allows playing games such as The Jumper, Hanoi Tower, and Four Horses.
- $[S_7] \rightarrow$  **Body Poses**. They are associated with the problem-solving process and represented by  $[S_7] = [joint\ hands,\ rub\ hands,\ stretch\ arms,\ put\ a\ hand\ on\ head,\ put\ a\ hand\ on\ chin,\ raise\ arms]$ .
- $[S_8] \rightarrow$  **Facial Gestures**. These are represented for the vector:  $[S_8] = [engagement,\ look\ up,\ look\ toward\ a\ point,\ close\ eyes,\ turn\ head,\ close\ left\ eye,\ close\ right\ eye,\ head\ flexion]$ .
- $[S_9] \rightarrow$  **Facial Emotions**.  $[S_9] = [happy,\ sad,\ anger,\ surprise,\ fear,\ disgust]$ .

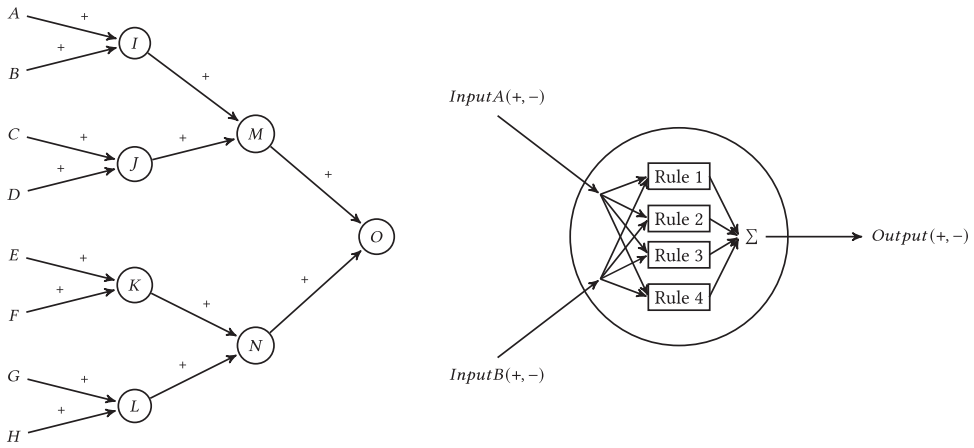


Fig. 2. Fuzzy Cognitive Binary Tree *FCBT*. (a) The Fuzzy Cognitive Binary Tree *FCBT* represents the expert knowledge through hierarchical relationships. Four features describes *FCBT*: nodes, arcs, arcs polarity, and sheet values. The inputs are: knowledge of the problem (A), loops in the trajectories (B), learner corrects the last movement (C), robot corrects the learner's movement (D), distance between current state and goal state (E), relevance of learners movement (F), branch by depth (G), and percentage of better continuous options made by learners (H). The nodes mean non-factual variables. The first four nodes describe divergent thinking processes necessary to solve problems: Representation (I), Flexibility (J), Comparison (K), Planning (L). The last two represent two fundamental skills for implementing the strategy: estimate options (M) and select the best option (N). Finally, level of the Mean-End Analysis skills (O). (b) The figure on the right side shows a basic description of the nodes. Each node uses the same fuzzy logic module. The attributes considered in each module have a valence to define their proportionality of contribution, either direct or inverse. Thus, the specific rule and variable for each valence case were defined (++ , +- , -+ , --).

As a result, the sensors consider the information associated with emotional and cognitive learner behavior. The following section explains how the data to feed the agent's beliefs were used to activate the robot's goals.

## 2.2 Beliefs Module

The theoretical foundation of the proposed architecture is the BDI agent model (Beliefs, Desires, Intentions) [8]; beliefs determine the assertive actions of the agent according to the scaffolding strategy and guide the solution of the problem [71, 77]. The beliefs are grouped into two features in HRS-EDU: cognitive and emotional. The cognitive features account for the operational knowledge of the problem and skills associated with the strategy [12]. The emotional features recognize the state of the variables (boredom, curiosity, comfort, anxiety) according to the flow theory [20]. The model implemented for each block is presented below.

- $B_1 \rightarrow$  **Strategic Knowledge**. Since abstract concepts represent some learner's actions, then HRS-EDU uses a Fuzzy Cognitive Binary Tree to evaluate these actions [54]. This method begins with an ontological representation of the variable  $x$  to be measured. The representation requires a hierarchical association between concepts to build a binary tree. The result of the method is the valuation of a variable  $x$  based on partial binary evaluations. See Figure 2.
- $B_2 \rightarrow$  **Operative Knowledge**. It means the learner's knowledge of the rules and the operators solve the problem.
- $B_3 \rightarrow$  **Learner's Trajectory**. The scaffolding principle is to mediate according to the learner's cognitive needs as defined by the learning problem. The problems proposed for

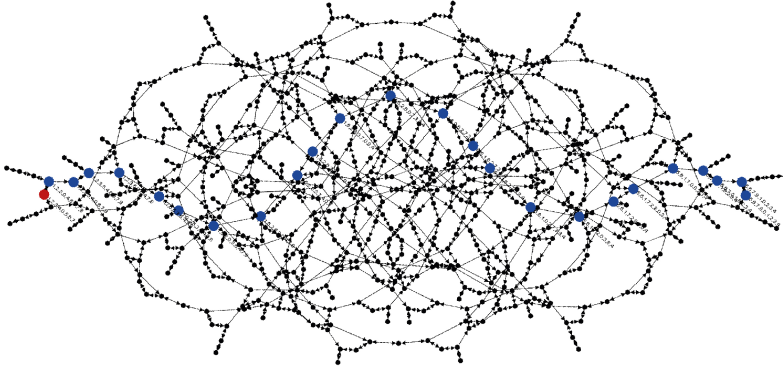


Fig. 3. The figure shows the problem space of the game The Jumper. The black nodes indicate the possible states, and the blue nodes mean the ideal path between the initial and goal state.

HRS-EDU are part of the category of well-defined problems. They have a problem space defined by the initial state, goal state, intermediate states, rules, and operators. The learning trajectory is evaluated on the graph to establish the cognitive needs. The Dijkstra algorithm was used for evaluated loops, best movements, and branches. These aspects were considered in the Fuzzy Cognitive Binary Tree mentioned above. See Figure 3.

- $B_5 \rightarrow$  **Database of the Learning Game.** Every learner's action is recognized over the problem-space. The number of options is related to the problem's structure and rules.
- $B_6 \rightarrow$  **Task State.** The block's position describes the learning trajectory, and it is necessary to trace the learning curve.
- $B_7 \rightarrow$  **Learner's Emotional State.** This is calculated by the sensor's information:  $S_7, S_8, S_9$ . The semiotic analysis identified 22 body gestures during the solution [11, 47, 62]. With a Kinect XBOX One, the body gestures were identified, and with a RealSense300 camera, the six emotional expressions were identified. The emotion recognition used Affdex SDK software, although MPT, HARK, CMU Sphinx, Synesketech, SENNA software were also evaluated. The K-Means technique for identifying emotional states was selected, although the EM and Canopy algorithms were evaluated with WEKA. The external validation of the clusters was carried out with 74 subjects. The data labeled by the subjects presented the following distribution: Boredom (18.87%), curiosity (43.98%), anxiety (23.22%), comfort (13.93%) [2].

In conclusion, the beliefs module of HRS-EDU has seven blocks to model the cognitive and emotional learner characteristics. This information defines the robot's behavior through the Desire-Intentions module.

### 2.3 Desires-Intentions Module

The practical reasoning model supports the design of HRS-EDU [8] and implemented in the BDI-BESA software [27]. Then, HRS-EDU considers the beliefs about the cognitive and emotional states of the students, desires to support the learning process according to the scaffolding strategy, and intentions to start the scaffolding process. The inputs of HRS-EDU consider facial expressions, body movements, and the task's state. Then, the *beliefs* module assesses the emotional state, knowledge, and skills that the learner has to solve the problem. The *desires* module uses the *beliefs* values to estimate the available goals to perform the scaffolding. The goal with the best score becomes an *intention*. According to the winning goal, the *intention* is performed using Baxter's actuators. The goals are grouped in the six sets described above and classified into one of the pyramid levels. Each level



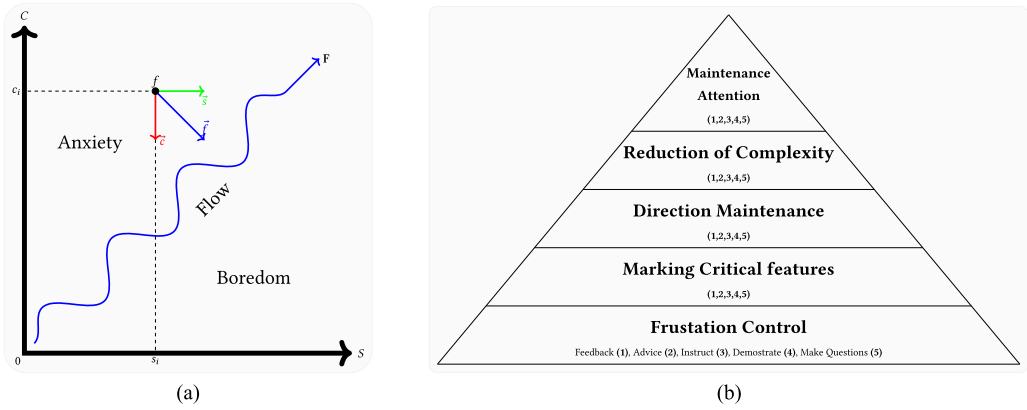


Fig. 4. Flow theory associates three variables: Challenge  $C$ , Skills  $S$ , and Emotions  $E$ . Challenge is linked to the subject's perception of the complexity of the problem. Skills mean the capacity to solve problems. Emotions are physiological responses to the relation between Skills and Challenge. The pyramid classifies the goals according to the robot's motivation into five levels related to the *scaffolding strategy*. The means (1, 2, 3, 4, 5) indicate the performance's alternative to achieve the goal.

considers four educational strategies (feedback, advice, instruct, demonstrate, make questions). (See Figure 4(b)). The values of the beliefs are updated while the human-robot interaction is ongoing. Consequently, the values of the contribution of the goals are updated too. The decision to keep or change the active goal depends on the student's emotions and cognitive state. (See Figure 4(a)).

- (1) *Goals to Foster Problem-Solving Skills*. The robot gives clues and prompts to support both the knowledge of the problem and strategy. The robot uses its emotional, cognitive, and physical skills in both cases.
- (2) *Goals to Achieve Cognitive Support*. HRS-EDU evaluates every change made by the learner according to the problem space. As a result, there is an assessment of the learner's movements. For instance, based on the rules, learners can make allowed movements, but they may not know the strategy. When the learner's movements are not relevant, the meta-cognitive goals (planning, supervision, and evaluation) are activated.
- (3) *Goals to Give Emotional Support*. Then, HRS-EDU fosters the flow state through three goals: robot controls the anxiety and boredom states through three goals: acknowledging success actions, acknowledging failures, and fostering confidence.
- (4) *Goals to Control the Challenge*. Being in the flow state is linked to maintaining a proper relation between skills and challenge. To variate the challenge, HRS-EDU increases or decreases the number of blocks to change the complexity of the task. For instance, robot advances or returns one or two states of the game.
- (5) *Goals to Give Life Signals*. HRS-EDU uses life signals to foster in learners the recognition of the robot as an agent involved in the learning process. The life signals shown by the agent are breathing, evoking thought, blinking, and whistling.
- (6) *Goals to Provide Immediate Support*. The scaffolding process involves immediate responses during critical events which risk the learning process. These goals give immediate support when the learner request help. The robot's support can either speak or arrange the game's blocks.

In HRS-EDU, the emotional behavior is based on the circumplex model of teacher emotions, and the axes are Control, and Affiliation [19]. As shown in Figure 5, the Cartesian plane describes

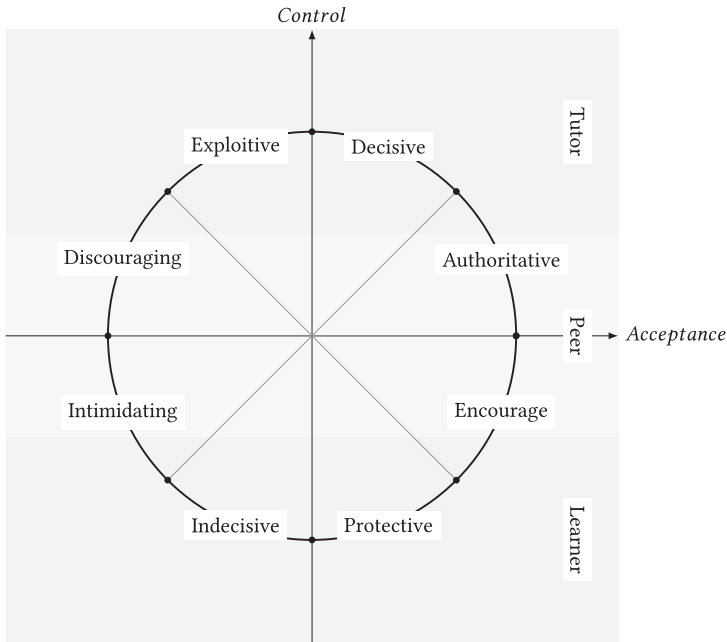


Fig. 5. The robot's roles are grounded in the circumplex model of principal-teacher interactions [7]. In HRS-EDU, the emotional expression model is based on [13] which describes a study about teachers' emotions. In learning contexts, emotions are a control mechanism. HRS-EDU has two variables to control how the robot presents the scaffolding. The variables are Acceptance and Control. The combination of both variables defines an emotional state associated with the robot's roles. The function relates the robot's actuators (fans, lights, movement speed, display) to the Acceptance and Control variables.

emotional attitudes as pleased, guiding, strict, encouraging, dissatisfied, doubtful, and interested. The Control axis defines the actions of emotional follow-up oriented to the feedback of the actions during learning. The Affiliation axis defines the degree of closeness between the learner's actions and the best movements to solve the game. The BDI-Goals are routines to control the robot's behavior. The goals were designed based on the scaffolding strategy and the flow theory. The robot's behavior results from the competition of goals; the winning goal takes control of Baxter until the goal is either finished or expropriated by another goal. HRS-EDU was implemented in BDI-BESA 3.0, a research product of the SIRP group of the Javeriana University. Additionally, the ROS operating system was used to integrate HRS-EDU with the anthropomorphic robot Baxter. The implementation of HRS-EDU was done in JAVA, the libraries of BDI-BESA, FuzzyLogic, JAVARosbridge, Graph Stream, Xbox ONE Toolbox were used.

### 3 RESEARCH METHOD

According to Figure 6, the project implemented the mixed-methods research strategy to evaluate the subjects' performance and the HRS-EDU architecture through Baxter Robot. The validation evaluates the design of HRS-EDU (Figure 1), which fosters the development of skills of the Mean-End Analysis strategy. This strategy is based on decomposing a problem into minor problems and try to solve each one to solve the main problem. In well-defined problems such as The Jumper, the strategy identifies the best movement of the blocks to achieve the goal. Since the robot's behavior



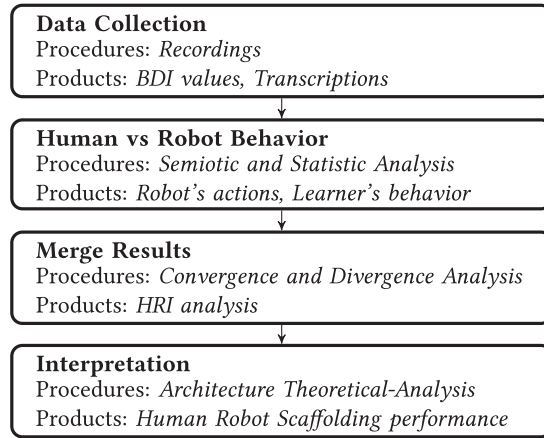


Fig. 6. Research Method. The Human-Robot Scaffolding strategy needs to analyze variables as the robot's behavior and the learning process. The quantitative data comes from the computer. The qualitative data comes from the video analysis of the experimentation. Finally, both datasets are mixed with analyzing how Baxter fosters the learning process.

results from the competition of the goals, then the validation considers the result of the robot's actions during the solution of the game. The game is appropriate for validation because it does not require prior knowledge, the rules are simple, and it requires scaffolding. The process was conducted with learners ranging from 10 to 14 years old. The validation did not consider a longitudinal study because of the schools' schedule and distance between the schools and the robotic laboratory. The researcher's participation was restricted from either the robot's mistakes or the subject's misunderstanding. The details of the validation process are explained below.

### 3.1 Task Description

The game to evaluate the architecture was *The Jumper*. The game has an initial state, goal state, rules, and operators to achieve the goal. The game has two sets of blocks, each with a different color. The goal is to exchange the position of both sets of blocks. The game has two rules: the blocks jump over the next empty position; this could be either to the right or left side, and the blocks can only jump over a block belonging to the other group of blocks. The game begins with the two sets of blocks separated, and the gamer needs to achieve the goal state through the movement of blocks based on the two rules already mentioned. Every movement of blocks is considered a change of state. See Figure 7.

### 3.2 Scenario

The Baxter robot was used to test HRS-EDU. Baxter is an anthropomorphic robot from Rethink Robotics. Baxter has two arms, each with a gripper, a camera, and an infrared sensor. Additionally, Baxter has an LCD screen to express emotions. To test HRS-EDU, Baxter was complemented with three deeper cameras, a microphone, and a speaker. The depth cameras were used to recognize the blocks' position, facial emotions, and body gestures. Between Baxter and the subject, there is a table. The research took place in a robotic laboratory Figure 8. At this location, subjects solved the problem while Baxter supported learning. The observation strategy involves two video cameras to focus on both the robot and the subject.

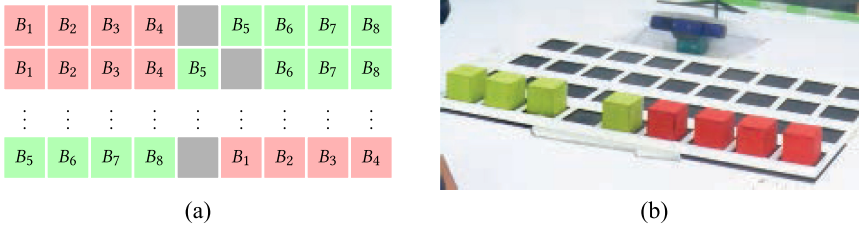


Fig. 7. Task description. The figure shows two features of the game (*changes of state, game board*). Figure (a) is a brief example of three states. Every row means a new state according to the two game's rules. The ideal path needs twenty-four changes to achieve the goal, but novices usually spend more than forty-five state changes to do it. Figure (b) shows the board game. Blocks are arranged in the first row of the board. Every change of state is in the same row.



Fig. 8. The research scenario is composed of the learner, Baxter, and the block of the game. Learners come from three different schools in Bogota City. They are in sixth and seventh grade. The robot used is Baxter. The Jumper is a transformation game, and it has a high dropout rate. A short video fragment that summarizes the research process is available in <https://youtu.be/qbohCjBIwYc>.

### 3.3 Variables and Data

The dependent variables are *skills, cognitive control, emotional state, challenges, life signals, role, support, and attention* these correspond to the set of goals proposed in HRS-EDU. The independent and intervening variables are *time and the HRS-EDU*.

Table 1. The four Variables Considered in the Interaction Were Emotions, Attention, Robot's Life Signals, and Actions

| Topic/Subject              | S1 | S2 | S3 | S4 | S5 | S6 | S7 | S8 | S9 | S10 | S11 | S12 | S13 | S14 | S15 | $\bar{X}$ |
|----------------------------|----|----|----|----|----|----|----|----|----|-----|-----|-----|-----|-----|-----|-----------|
| <b>Emotions</b>            |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |           |
| Boredom                    | 4  |    |    |    |    |    | 7  |    |    |     |     |     |     |     |     | 0.73      |
| Anxiety                    | 1  |    |    |    |    |    |    |    |    |     |     |     |     |     |     | 0.06      |
| Comfort                    | 87 | 57 | 38 | 63 | 62 | 73 | 9  | 88 | 61 | 41  | 47  | 46  | 53  | 19  | 57  | 53.4      |
| Curiosity                  |    |    |    |    |    |    | 1  |    |    |     |     |     |     |     |     | 0.06      |
| Apathy                     |    |    |    |    |    |    | 1  |    |    |     |     |     |     |     |     | 0.06      |
| <b>Attention</b>           |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |           |
| External Stimulus          | 3  |    |    |    |    |    | 3  |    |    |     |     |     |     |     |     | 0.4       |
| Researcher                 | 3  |    |    |    |    | 5  | 1  | 1  |    |     |     | 1   | 2   | 2   | 1   | 1.06      |
| Baxter's Movements         | 6  | 1  | 3  | 5  | 1  | 4  | 3  | 2  | 11 |     |     | 5   |     | 3   | 2   | 3.06      |
| Baxter' Screen             | 24 | 2  | 6  | 7  | 11 | 14 | 2  | 13 | 6  |     | 1   | 3   | 6   | 2   | 12  | 7.26      |
| Game Board                 | 56 | 54 | 29 | 49 | 46 | 45 | 8  | 67 | 44 | 41  | 46  | 40  | 45  | 11  | 44  | 41.6      |
| <b>Actions</b>             |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |           |
| Baxter changes the state   | 9  | 1  | 1  | 2  | 1  | 4  | 2  | 7  | 1  | 1   |     | 3   |     | 2   | 3   | 2.46      |
| Baxter gives clues         | 43 | 20 | 10 | 15 | 12 | 17 | 6  | 20 | 12 | 2   | 3   | 17  | 7   | 3   | 11  | 13.2      |
| Start of the problem       | 1  | 1  | 1  | 1  | 1  | 2  | 1  | 1  | 1  | 1   | 1   | 1   | 1   | 1   | 1   | 1.06      |
| Subject changes the state  | 39 | 35 | 26 | 43 | 44 | 45 | 8  | 56 | 47 | 37  | 42  | 25  | 42  | 10  | 38  | 35.8      |
| <b>Robot's life signal</b> |    |    |    |    |    |    |    |    |    |     |     |     |     |     |     |           |
| Evoke Thinking             | 14 | 8  | 2  | 1  | 3  | 3  |    | 6  | 1  |     |     | 4   | 1   |     | 1   | 2.93      |
| Blink                      | 2  | 3  |    |    |    |    |    |    |    |     |     |     |     |     |     | 0.33      |
| Breathe                    |    |    |    |    |    | 1  |    | 1  | 1  |     |     | 2   |     | 1   | 1   | 0.46      |
| Whistle                    | 5  |    |    |    |    |    |    | 2  |    |     |     |     |     |     | 2   | 0.6       |

The emotions are grounded in the psychological theory of flow. Attention means where the student is looking, either the game, robot, or researcher. The robot's life signals describe the robot's actions to evoke life. Actions mean the explicit actions of the subject or Baxter to solve the problem.

- *Skills.* Knowledge of the subject to make decisions based on the Mean-End Analysis strategy.
- *Actions.* They are divided into two groups: the subject's and the robot's actions. The subject's action is to move the blocks. The robot's actions are to move the blocks and give verbal support.
- *Emotional State.* The researcher's perception of the learner's emotions is based on the flow theory.
- *Life Signals.* Robot's actions evoke life as a human.
- *Attention.* The researcher's perception of the subject's attention could be oriented to either problem's board, robot, researcher, or external stimulus.

The procedure to collect the data was to bring the subjects to the robotic laboratory, start the recording, explain the research procedure, the subject solving the problem, and stop the recording. The data were grouped in the recordings, architecture's internal data, and trajectory used by learners to solve the problem. Figure 9 shows two examples where Baxter interacts with the subjects. The robot's behavior considered three roles: tutor, peer, and learner. The robot's roles were designed according to the scaffolding theory. However, the robot's roles were performed implicitly; during the interaction, the robot did not say things such as "I am a tutor, I am a partner, I am a novice." These roles are based on the model of interpersonal teacher behavior. The interaction establishes a condition of cognitive and emotional dependence. Emotional dependence means that subjects expect feedback signals during the learning process. This condition is evident between tutor-learner, learner-learner, learner-tutor. The received signals influence the learning trajectory. In teachers, emotions are a function of beliefs, desires, intentions, identity, and teaching objectives [13]. See Figure 5. Moreover, the subject's behavior was considered to analyze the human-robot scaffolding phenomenon. The research had a Convergent-Parallel Approach. Figure 6 is



Fig. 9. The physical resources used by Baxter are grippers, speakers, lights, fans, and a display. Two different examples in which Baxter does scaffolding to foster the problem-solving skills are presented. The materialization of the robot's roles is a combination of various resources. For instance, the role of being pleased involves mixing a pleasant face, green lights, low fan speed, slow movement speed, and a gentle emotional tone of expression.

divided into four stages: Data Collection, Analysis of Human vs Robot behavior, Merge Results and Interpretation.

#### 4 RESULTS AND DISCUSSIONS

The most important characteristics are:

- *Trajectory.* Figure 3 shows the problem space for the game. Now, Figure 10 is a heat map that summarizes the trajectories of solution of the problem made by the subjects with the Baxter mediation and without the Baxter mediation. The short path for the game is highlighted with a gray line. As a result of Baxter's support, subjects described shorter trajectories to solve the problem. In contrast, without Baxter's support, subjects took longer to solve the problem or randomly navigated the problem space.
- *Cycles.* In Figure 10, the color and size of nodes mean the number of times a node was visited. For instance, a graph with all nodes with the same color means the nodes were visited once, but a graph with different colors and different size nodes means the nodes were visited at least twice. As shown in Figure 10, when learners solved the problem, the graphs on the left side had more cycles than the graphs on the right side. The difference in the number of cycles was caused by Baxter's support given through hints or movements of the blocks.
- *Development of Skills.* Skills mean correctly applying the Means-Ends analysis strategy to solve the problem. The goals proposed in the architecture were designed to promote the subject's skills. Baxter promoted learning through block movements, verbal aids, and emotional support. During the problem-solving process, the subject's attention had different focuses, such as the arms, screen, and speech movement. (See Table 1.) The first meet was

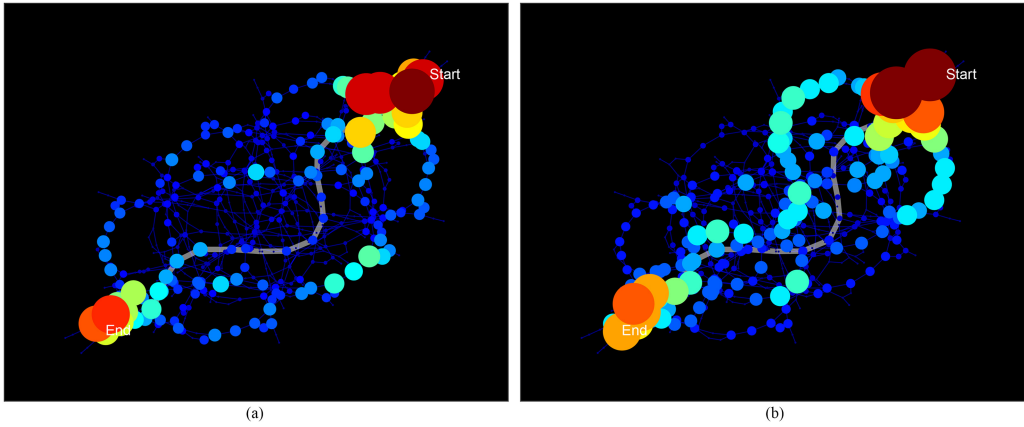


Fig. 10. The figure compares the graphs with and without Baxter's support. In both figures, the colors change between blue and red. Blue indicates that the nodes were not visited. Red mean a high amount of visits done by the learners. Figure 10(a) indicates the weak presence of the problem-solution path. Most of the movements made by the subjects are in the initial and final states of the game problem space. In contrast, Figure 10(b) indicates a marked problem-solving trajectory. The difference between the two figures verifies the positive contribution of the robot's behavior to the problem's solution.

affected by the novel effect, but the attention to Baxter decreased while the subjects were focused on the problem-solving process. Subjects showed apathy towards Baxter's actions when subjects had already deciphered Baxter's actions during problem-solving. According to the analysis of the videos, the subjects showed interest in help only when they were confused. When the subjects longer than required Baxter's help time, they showed anxiety to regain control of the game solution.

- *Emotional Control.* HRS-EDU triggers goals according to the scaffolding strategy. The actions were designed to foster the flow state while subjects solved the problem. As a result of the research, the learner's predominant expression was comfort. (See Table 1.) However, some learners expressed boredom due to the test time, and subjects showed anxiety when they needed to wait until the robot finished its movements but knew the solution. Indeed, boredom and anxiety were associated with the human-robot interaction process instead of the problem-solving process.
- *Challenge Control.* HRS-EDU triggers goals to move in forward or backward states of the problem. According to the data analysis, the robot's actions were assertive. The learners paid attention to the cues or hints made by the robot. If learners did not know the strategy, they used the time when the robot was moving to think about the solving process. Table 1 shows main data about the learner's attention. According to the data, Baxter's actions were fewer than the learner's interventions, and the attention to Baxter's movements was greater when the subject requested help than when the robot decided to give support.
- *Life Signals.* In HRS-EDU, the life signals objective is evoking life. However, during the experiments, the subjects sometimes acknowledged its signals as Baxter's regulation of the cognitive and emotional processes. Life signals are not considered in the scaffolding strategy, but in architecture, they contribute to the recognition of Baxter as a collaborative agent. When it was activated, two types of actions were prominent: subjects looked or did not look to the robot's screen due to the problem demanding spatial orientation. Thus, the subjects kept their eyes on the board while they finished the movements. However, the subjects looked at the screen when they intended to move or plan without manipulation.



- *Support.* It is a high-priority goal to advance the problem. The physical interventions of the robot reflected in the movements of the blocks generated a visual and spatial representation of the problem that facilitated understanding actions according to the scaffolding strategy. The pauses generated by Baxter's movements created opportunities to think about the problem and continued with greater security after Baxter's intervention. On the other hand, emotional states of anxiety or boredom before or after the robot's intervention were not shown; the attitude of comfort was maintained.

As a result, the *BDI-Goals* and the design of the plans contributed to the overall objective of the HRS-EDU. The actions' relevance measures how appropriate the movements of the subjects when they must choose between different possibilities to change states to solve the problem.

## 5 CONCLUSIONS

Pedagogical and psychological theories were used to integrate three aspects: the five sets of goals (skill development, emotional control, challenge control, life signs, cognitive control), the goals hierarchy (frustration control, highlighting critical conditions, maintain direction, reduce degrees of freedom, maintain attention) and the means of intervention (feedback, advice, instruction, demonstration, questions). The novel articulation of the three aspects mentioned above promotes an assertive behavior of the robot while assisting the subject during learning. The subjects paid attention to the instructions for solving the problem. The instructions are given by Baxter corresponded to the cognitive and emotional needs of the subject. When the learners were confused and asked for help, Baxter's interventions were assertive thanks to two actions: unblocking the problem solution and generating a delay to think about the solution based on the robot's strategy.

Critical problems are joined into four aspects. First, recognizing the pose and emotional expression requires closeness to the cameras, making the setting for the experiment even more artificial. Second, the physical support given by Baxter was helpful while the learner did not know the solution of the problem, but by the time the learner discovered the solution, Baxter's movements caused anxiety. Third, the emotional expressions of Baxter were only considered while the subject was thinking about the problem, and his attention was not focused on the game board. Finally, the occlusion generated by manipulating the blocks makes their recognition difficult. This last condition restricts the selection of games that require blocks and different forms of spatial organization.

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